

What Does Drama Sound Like?

Sound Analysis of GerDraCor

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Abstract. This methodological pilot study examines how the computational method for sound recognition and classification developed for German-language prose by Guhr (2026) can be adapted to dramatic texts. Using the German Drama Corpus (GerDraCor, Fischer et al. 2019), we manually annotate twelve plays, apply a rule-based heuristic that completes elliptical stage directions, and fine-tune a pretrained German-language BERT model for automated sound annotation. Cross-validation experiments show that prose data supplies genre-general sound event recognition while the annotated plays supply drama-specific structural knowledge, but only once their elliptical sound indications have been assimilated to the syntax of prose: Added in their original form the plays lower performance, added in augmented form they raise exact-span detection. Applying the model to 650 GerDraCor plays, we find a statistically significant diachronic increase in Sound Event Density (SE_D), partly mediated by the historical expansion of stage directions and specific to character sound, as well as significant differences in sound use across literary movements, and author-related patterns.

1. Introduction

Shouts, whispers, and offstage crashes – sounds shape dramatic texts far beyond spoken dialogue. Although unmediated dramatic speech dominates the genre, auditory elements also appear in stage directions, through expressive qualifiers, or as self- or other-assigned sound descriptions within the dialogue itself.

In a diachronic sound analysis of the German Drama Corpus (GerDraCor, Fischer et al. 2019), this paper investigates sound usage across centuries. Sound is hereby understood as the explicit mentioning of anything that is / would be audible, and dramatic texts can represent them in different ways: On the one hand, there is the dramatic speech, which makes up the vast majority of a play as the primary text (Pfister 2001, 6). It often contains attributed speech descriptions of the speaker or other characters. For instance, in Sachs's *Eulenspiegel*, there are character sound assignments, such as the self-assigned sound in Example 1a and the other-assigned sound in Example 1b. On the other hand, the stage directions in the secondary text also indicate sounds, such as in Example 1c, or they add information on the desired realization of a dialogue related to sound and its quality, such as in Example 1d. In this investigation, we focus on these explicit markers of sound in dramatic texts and make use of existing research originally developed for

the operationalization of fictional sounds in narrative prose (Guhr 2026), distinguishing between character and ambient sounds based on their origin (sound source identifiable as character(s) or as part of the identifiable or non-identifiable background sounds).

Example 1a. “Darum ich euch sehr fleißig **bitt**”

“I kindly **request** that you do so”.¹ (*Eulenspiegel*, Sachs)

Example 1b. “Wie du jetzt selber hast **gered’t**”

“As you have **stated** yourself”. (*Eulenspiegel*, Sachs)

Example 1c. “<stage>Man hört noch einen **Schuß**.</stage>”

“<stage>Another **shot** is heard</stage>”. (*Die Räuber*, Schiller)

Example 1d. “ORSINA <stage>**heftig**.</stage> Nicht gelesen?”

“ORSINA <stage>**fiercely**</stage> Didn’t read?”. (*Emilia Galotti*, Lessing)

Concretely, we investigate two research questions on GerDraCor: (RQ1) First, we wonder how an annotation and operationalization workflow originally developed for sound in prose can be adapted for application to dramatic texts, and what genre-specific adjustments to the annotation guidelines, e.g., regarding the treatment of stage directions or sound descriptions embedded in character speech, are necessary to maintain annotation reliability and conceptual consistency with the prose-based scheme. The absence of a narrator in drama, together with dramatic features such as explicit and implicit stage directions or the particular strategy of teichoscopy, can be expected to affect how ambient sound in particular is represented and integrated into the text, raising the question of whether an annotation scheme designed around narratorial mediation transfers to a genre that largely lacks it. (H1.1) We expect that reusing the prose-trained sound annotation model alone is insufficient for dramatic texts: Applying a model fine-tuned exclusively on sound-annotated prose (Guhr 2026) to GerDraCor plays would perform poorly because the model has not learned the structural conventions of dramatic texts, containing speech prefixes, stage directions, and the GerDraCor-specific XML TEI encoding. We therefore hypothesize that further fine-tuning the model on sound-annotated plays is necessary to adapt it to the dramatic text structure, while the sound-annotated prose data continues to contribute valuable data for recognizing sound event units, independent of genre. (H1.2) Accordingly, we assume that combining drama-specific structural adaptation with increasing amounts of prose-based sound event examples improves performance on dramatic texts, reflecting a division of learning between genre-specific structural learning and genre-general sound event recognition. However, we still expect performance to vary considerably across individual plays and sound types, reflecting the corpus’s structural diversity and the persistent challenges of cross-genre transfer.

(RQ2) Second, we ask how sound is used in plays across centuries and different literary movements. Analyzing the automated sound annotations, we perform a frequency-based comparison of sounds in the different plays, filtered by time period, literary movement, and author, to identify patterns and differences in sound use diachronically. (H2.1) We hypothesize that plays published in the 20th century will provide more sound representations compared to earlier periods, reflecting the rise of experimental theater traditions, though we also consider whether this increase may partly mirror the historical

1. German quotes were translated into English, supported by the software DeepL Translator (DeepL SE 2026).

expansion of stage directions (Trilcke et al. 2020). (H2.2) Finally, we anticipate that certain authors and literary movements, in particular Naturalism and Expressionism, will demonstrate distinct sound usage patterns reflecting author- and movement-related particularities of sound representation in GerDraCor.

2. Computational Approaches to Sound in Drama

Computational drama analysis is a well-established branch of computational literary studies, addressing, e.g., the analysis of authorship and style (Agapitos and Cranenburgh 2024; Matthews 1993) or genre (Szemes and Nagy 2025), which make use of drama-specific features such as the attribution of speech to characters. Drama-specific research has focused on character properties, including character type (Krautter and Pagel 2024; Krautter et al. 2020, 2022), co-presence on stage (Trilcke et al. 2024), and emotions (Dennerlein et al. 2024), as well as on the utterances of individual characters with respect to style (Vishnubhotla et al. 2019) and topic (Karsdorp et al. 2015; Reiter and Willand 2023).

Sound in literary texts has received limited systematic attention. In prose, studies by Picker (2003), Glotova (2021), Molyneux (2022), and Foley (2023) have examined sound from a literary-historical perspective, but these analyses have typically relied on close reading of single texts. Guhr (2026) offers the first computational operationalization of sound events in literary prose, enabling sound analysis at corpus scale. Building on the event concept developed by Gius and Vauth (2022), she defines fictional sound as follows:

A fictional sound is a sound event with asserted *realis* in the fiction that is explicitly represented in the narrative by the presence of sound words, which are sound-representing content words used in the narrative to provide information about the fictional soundscape (e.g., ‘Stimme’/‘voice’, ‘rattern’/‘to rattle’, ‘bellen’/‘to bark’). The realization of fictional sounds can be represented in more detail by sound event qualities like loudness, which carry semantic information about how a fictional sound was realized or perceived.

For diachronic and synchronic comparison, she proposes analyzing the density of sound words (SW_D) and sound events (SE_D) across texts and presents a method for dictionary-based loudness labeling that could, in principle, be adapted for dramatic texts, though the vocabulary range of GerDraCor, spanning four centuries and frequently including vernacular speech and unconventional spelling, renders this approach difficult to apply directly and remains a challenge for future research.

Specifically in drama, ambient and character sounds have, to our knowledge, not yet been systematically investigated, including not only the presence of sound events, but also their manner, quality, and intensity as encoded in stage directions beyond dialogue. Existing scholarship has approached dramatic sound primarily from performative and semiotic angles. Performance-oriented research has explored the physicality of speech and the spatial dynamics of stage interaction, emphasizing perception and mediation as central to understanding sound in theater (see Hillebrandt (2018, 124),

Mildorf (2019, 293), and, generally, Kendrick (2017) and Smith (1999)). From a semi-otic perspective, Aston and Savona (1991, 84–87) offer frameworks for distinguishing the communicative functions of sound within theatrical performance. However, both approaches presuppose a close relationship between the written dramatic text and its theatrical realization that is difficult to operationalize computationally across a diachronic corpus. Dramatic texts present a particular challenge for computational text analysis: Confined to written form, they encode vocal and performative elements that are only fully realized in performance. We therefore build on Guhr (2026)’s operationalization, which is grounded in the written text, computationally tractable, and applicable for corpus-scale approaches.

3. Data and Method

For our investigation of sound in drama, we use the German Drama Corpus (GerDraCor), provided by DraCor (Fischer et al. 2019). The corpus spans approximately four hundred years, from around 1550 to 1950, with most plays having a publication date between 1750 and 1950 ($n=679$).² Applying the German BERT model (Chan et al. 2020) fine-tuned for sound prediction on prose texts by Guhr (2026) directly to GerDraCor plays transfers only partially: The model reaches a token macro F1 of 0.39 and an E-F1 of 0.35 across the twelve manually annotated plays, well below its performance on prose (> 0.7), and its errors are characteristic of a model that has not learned the structural conventions of dramatic texts, such as speaker prefixes, stage directions, and the elliptical constructions typical of the secondary text (see error analysis in subsection 3.3). The prose model recognizes sound events expressed as complete verbal phrases, as they are in prose, but does not account for the ways drama differs structurally. We therefore manually annotated plays from the GerDraCor to create drama-specific training data (see the manual annotation workflow visualized in Figure 1).

3.1 Manual Annotation of Sound in Drama

Manual Annotation Workflow. Following the guidelines developed by Guhr (2026), the annotation proceeds in two steps: (1) The text is first segmented into syntactic phrases containing a semantically identifiable sound word representing a sound event, then (2) each detected sound event is classified as either ‘character sound’ or ‘ambient sound’ according to its source – a sound-producing character in the former case, or, in an *ex negativo* decision, an ambient sound where no identifiable character can be determined as the source, in the latter. Character sounds produced by characters speaking on stage are the default case in drama, marked by the combination of speech prefixes and the respective dialogue in the primary text (Pfister 2001, 23). Our analysis, however, does not address the frequency of characters speaking in plays, which has already been covered by several studies (e.g., Fischer et al. (2018) and Reiter (2021)).

Pfister (2001, 13–14) distinguishes prose and drama by their presentation structures: Prose usually employs a mediating narrator who shapes the story and its interpretation, whereas drama communicates directly between characters and audience. Lacking a narrator, drama distributes sound information across two channels, the performative

2. As of the update on April 9, 2026, GerDraCor contains 769 plays.

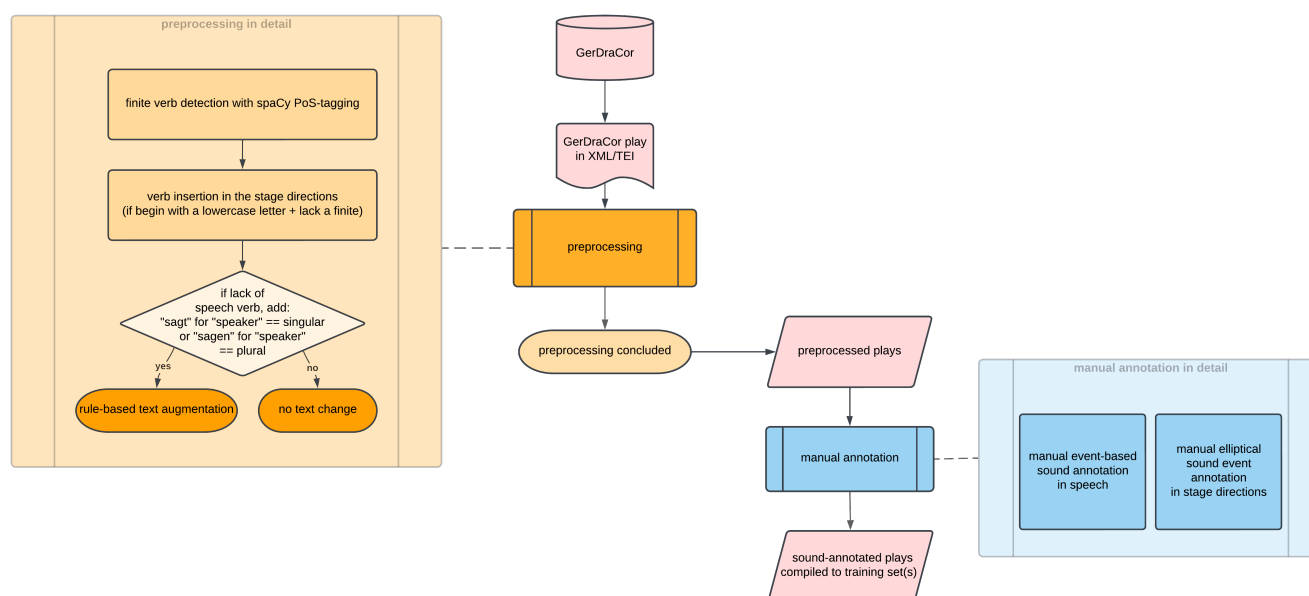


Figure 1: Manual annotation workflow for sound annotation of GerDraCor plays, covering preprocessing and manual annotation process. Node colors indicate step type: orange = preprocessing and postprocessing steps; blue = manual annotation steps; rose = data. The detail box on the left elaborates the preprocessing (stage direction augmentation via rule-based verb insertion).

dialogue of the primary text and the multimodal instructions of the stage directions in the secondary text. Our analysis focuses on the explicit indications of sound that go beyond dramatic speech itself, for instance the additional information conveyed by the combination of a stated speaker and a stage direction indicating how an utterance is to be realized with respect to sound markers and sound qualities, such as “ORSINA *fiercely*” in Example 1d.

This distribution across primary and secondary text complicates event-based sound detection in two ways: (1) First, the verbal phrase constituting the sound event is interrupted by the XML TEI elements <speaker>, <stage>, and paragraph <p> or line <l>, so that the combined speech-prefix/stage-direction/dialogue unit cannot be annotated or automatically recognized as a single sound event span. (2) Second, stage directions indicating sound qualities frequently rely on ellipsis: Rather than including an explicit sound word, they offer adjectives or adverbs that describe the manner of speech delivery without encoding a complete verbal phrase, and since the *verbum dicendi* is missing, as in Example 1d, sound word-sensitive recognition fails. Dramatic texts therefore do not always provide complete sound events, even where sound is indicated.

The absence of a narrator has another consequence, which concerns ambient sound in particular. Where prose can report environmental or off-stage sound through narratorial mediation, drama must either place it in a stage direction, as in Example 1d, or have a character report it. The latter strategy, *teichoscopy*, integrates auditory elements from beyond the stage into the dramatic world through description mediated by characters on stage. In Schiller’s *Maria Stuart* (see Example 2), Leicester, unable to witness the execution, instead reports what he hears from below, “schluchzen” (‘sobbing’), “weinen” (‘weeping’), and “dumpfes Getöse von Stimmen” (‘muffled roar of voices’), through a series of references to ambient and character sounds (Pfister 2001, 206, 270).

Example 2. <div type="scene"> <head>Zehnter Auftritt</head>
 <stage>Leicester allein zurückbleibend .</stage>
 <sp who="leicester">
 <l>Ich lebe noch! Ich trag es, noch zu leben!</l> [...]
 <l>Ich kann, ich kann das Schreckliche nicht schauen,</l>
 <l>Kann sie nicht sterben sehen – **Horch!** Was war das?</l>
 <l>Sie sind schon unten – Unter meinen Füßen</l>
 <l>Bereitet sich das fürchterliche Werk.</l>
 <l><ambient_sound>Ich höre Stimmen</ambient_sound> – Fort! Hinweg! Hinweg!</l> [...]
 <l>Muß ich anhören, was mir anzuschauen graut?</l>
 <l><character_sound>Die Stimme des Dechanten – Er ermahnet sie</character_sound> – </l>
 <l>– <character_sound>Sie unterbricht ihn</character_sound> – **Horch!** – <character_sound>Laut
 betet sie</character_sound> – </l>
 <l><character_sound>Mit fester Stimme</character_sound> – <ambient_sound>Es wird
 still</ambient_sound> – <ambient_sound>Ganz still</ambient_sound>!</l>
 <l><ambient_sound>Nur schluchzen hör ich</ambient_sound>, <ambient_sound>und die
 Weiber weinen</ambient_sound> – </l>
 <l>Sie wird entkleidet – **Horch!** Der Schemel wird</l>
 <l>Gerückt – Sie kniet aufs Kissen – legt das Haupt –</l> </sp>
 <stage><character_sound>Nachdem er die letzten Worte mit steigender Angst gesprochen</char-
 acter_sound> , und eine Weile inne gehalten, sieht man ihn plötzlich mit einer zuckenden Bewe-
 gung zusammenfahren, und ohnmächtig niedersinken, <ambient_sound>zugleich erschallt von
 unten herauf ein dumpfes Getöse von Stimmen</ambient_sound>, <ambient_sound>welches
 lange forthallt</ambient_sound>.</stage>

<div type="scene"> <head>Scene 10</head>
 <stage>Leicester remains alone.</stage>
 <sp who="leicester">
 <l>I am still alive! I can bear to still be alive!</l> [...]
 <l>I cannot, I cannot bear to look upon the horror,</l>
 <l>I cannot watch her die – **Listen!** What was that?</l>
 <l>They are already down below – beneath my feet</l>
 <l>The dreadful deed is being prepared.</l>
 <l><ambient_sound>I hear voices</ambient_sound> – Away! Go away! Go away!</l>
 [...]
 <l>Must I listen to what I dread to see?</l>
 <l><character_sound>The dean's voice – He admonishes her</character_sound> – </l>
 <l>– <character_sound>She interrupts him</character_sound> – **Listen!** – <charac-
 ter_sound>She prays aloud</character_sound> – </l>
 <l><character_sound>With a firm voice</character_sound> – <ambient_sound>It grows
 quiet</ambient_sound> – <ambient_sound>Completely quiet</ambient_sound>!</l>
 <l><ambient_sound>I hear only sobbing</ambient_sound>, <ambient_sound>and the
 women are weeping</ambient_sound> – </l>
 <l>She is undressed – **Listen!** The stool is</l>
 <l>Moved – She kneels on the pillow – lays her head –</l> </sp>
 <stage><character_sound>After uttering his final words with growing fear</character_sound>
 and pausing for a moment, he suddenly jerks in a startled movement and collapses unconscious,
 <ambient_sound>at the same time, a muffled roar of voices rises from below</ambient_sound>,
 <ambient_sound>which echoes for a long time</ambient_sound>.</stage>
 (Maria Stuart, Schiller)

While the sound description in the teichoscopy (Example 2) presents a particular, extreme case of mediated sound description through dialogue, Pfister (2001, 16) refers to speech-acts “impl[ying] an action carried out by either the speaker himself or one of the other figures” that can also “refer to the visual and acoustic context” (Pfister 2001, 15), such as in Example 1a, 1b, and Example 1c. However, teichoscopy as well as the action-referring speech acts can also be found in dialogues given in prose texts, building a bridge between prose and drama.

Heuristic Augmentation. To make these different sound indicators processable for both manual and automated annotation, we applied a heuristic that augments stage directions as a preprocessing step before annotation or prediction (see Figure 1).

(1) Stage directions following speaker indications <speaker> that begin with a lower-case letter and lack a finite verb are treated as the primary candidates for augmentation, because they are characteristic of the elliptical constructions typical of dramatic secondary text. They usually consist of short adjectival or adverbial phrases such as *sehr ruhig* (‘very calmly’) that describe the manner of an utterance without encoding the speech act itself, which is what makes finite-verb detection a usable filter. Based on part-of-speech tagging with spaCy’s German language model (de_core_news_sm, Montani et al. (2022)), the algorithm identifies finite verbs within each stage direction. If none is detected, the speech verb *sagen* (‘to say’) is inserted, conjugated as *sagt* (singular) or *sagen* (plural) according to the number of speaker identifiers in the who attribute of the enclosing <sp> element (see the original in Example 3 and the augmented version in Example 4, both taken from Hauptmann’s *Die Ratten*). Where the direction does describe a manner of speaking, this restores a syntactically coherent verbal phrase without altering the original meaning.

Example 3. <sp who=“nathanael_jettel”> <speaker>Nathanael Jettel,</speaker>
<stage>sehr ruhig.</stage><p>Könnten Sie mir nicht sagen, Direktor, wer Ihnen in Gottes Namen auf die Krawatte getreten hat?</p>

<sp who=“nathanael_jettel”> <speaker>Nathanael Jettel,</speaker>
<stage>very calmly.</stage><p>Couldn’t you tell me, Director, who on earth stepped on your tie?</p> </sp> (*Die Ratten*, Hauptmann)

Example 4. <sp who=“nathanael_jettel”> <speaker>Nathanael Jettel,</speaker>
<stage>**sagt** sehr ruhig.</stage> <p>Könnten Sie mir nicht sagen, Direktor, wer Ihnen in Gottes Namen auf die Krawatte getreten hat?</p> </sp>

<sp who=“nathanael_jettel”> <speaker>Nathanael Jettel,</speaker>
<stage>**says** very calmly.</stage> <p>Couldn’t you tell me, Director, who on earth stepped on your tie?</p> </sp> (*Die Ratten*, Hauptmann)

(2) In the XML TEI encoding conventions of GerDraCor, stage directions that form syntactically complete structures, including nominalized constructions, imperative forms, or verb-less but complete phrases, tend to begin with a capital letter, whereas purely elliptical manner adverbials tend not to. Capitalization therefore serves as a conservative filter that avoids inserting *sagt/sagen* where it would be semantically or grammatically inappropriate. One-word directions beginning with a capital letter are consequently not detected as candidates, a loss of coverage we accept in exchange for a lower risk of distorting the text. Looking through the manually annotated subcorpus, we

recognized that the applicability of the heuristic is unevenly distributed. In the earliest of our annotated plays, Ayres's *Fassnachtspiel* (1605), stage directions are formulaic and name the speech act explicitly, as in Example 5. The *verbum dicendi* there is also in Ayres's original. The direction already contains a finite verb and therefore is not a candidate for augmentation. What the heuristic targets is the elliptical manner adverbial of the *sehr ruhig* type, and plays that do not use it are left untouched.

This unevenness, however, does not follow publication date. Across the twelve annotated plays the augmentation adds 81 sound events, but the gain is concentrated in Lessing's *Emilia Galotti* (+30) and Ebner-Eschenbach's *Die Veilchen* (+32) rather than in the later plays: Borchert's *Draußen vor der Tür* (1947) gains none and Dovsky's *Mona Lisa* (1915) one, and the correlation between publication year and augmentation gain is not significant (Spearman's $\rho = -0.30$, $p = 0.34$). The heuristic thus responds to an encoding habit that varies by author rather than to a historical convention. This matters for the diachronic analysis in section 4: Since the augmentation can only add character sound events, a systematic bias toward later plays would have inflated precisely the trend we report, and we find no such bias.

Example 5. <speaker>Eva</speaker>

<stage>geht ein <character_sound>vnd sagt zornig</character_sound>.</stage>

<l>Junckherr, wo wilt so frü hinauß?</l>

<speaker>Eva</speaker>

<stage>enters <character_sound> and says angrily</character_sound>.</stage>

<l>Young man, where are you going so early?</l>

(*Fassnachtspiel*, Ayres)

To address cases not covered by the heuristic, we extended the annotation guidelines to cover reduced sound events in elliptical stage directions, such as one-word directions or directions lacking a full verbal phrase. Furthermore, during the manual annotation of the 12 augmented plays, the inserted verbs were reviewed, and those that did not correspond to a realized speech act, whether grammatically ill-formed or attached to a stage action rather than an utterance, were removed. This correction step applies to the training data only: The plays augmented for corpus-wide prediction receive the automatic heuristic without manual review (see section 4).

Example 6. <sp who="nerissa"> <speaker>NERISSA.</speaker>

<lg> <l>O Himmel! Muß ich dieses noch erfahren,</l> <l>Muß ich des Mörders Antlitz doch noch sehn!</l>

<l>Die Stimme die mir werth ist wieder hören</l>

<l>Und denken, <character_sound>daß sie jenes harte Wort</character_sound>,</l>

<l><character_sound>Das grause Todes-Urtheil ausgesprochen</character_sound>,</l>

<l>Das mich von seiner Liebe immer trennt.</l> </lg> </sp>

<sp who="nerissa"> <speaker>NERISSA.</speaker>

<lg> <l>Oh heavens! Must I endure this as well,</l> <l>Must I still see the murderer's face!</l>

<l>Hear that voice I hold so dear once more</l>

<l>And think, <character_sound>that it has spoken that harsh word</character_sound>,</l>

<l><character_sound>That dreadful death sentence</character_sound>,</l>

<l>That forever separates me from his love.</l> </lg> </sp> (*Udohla*, Günderrode)

A further challenge arises in plays written in verse, where the segmentation into verbal phrases conflicts with the line structure. In Example 6, the sound event phrase is interrupted by the <l> element, and since an XML element cannot span across the boundary of another, we resolved these cases through double annotation, which means that one sound event is represented by two annotation spans.

Despite these structural challenges, annotation is guided primarily by the presence of semantically identifiable sound words, and the character/ambient classification follows directly from whether an identifiable sound-producing character is given. The decision is therefore grounded in explicit textual evidence rather than subjective inference, which makes the task less sensitive to systematic disagreement than, for instance, sentiment or emotion annotation. Guhr (2026) reports an inter-annotator agreement of > 0.8 Krippendorff’s alpha for sound event detection and classification on prose, which we take as a basis for our reasoning that the underlying scheme is reliably applicable by a single trained annotator. All plays in this study were accordingly annotated by one expert annotator with experience in literary sound annotation. Two qualifications apply: Agreement was established on prose and therefore covers neither the drama-specific extensions to the guidelines introduced above, namely reduced sound events in elliptical stage directions and double annotation across verse lines, nor the review of heuristically inserted speech verbs. These are the decisions on which annotators would be most likely to diverge, and establishing agreement for them on dramatic texts remains open for future work.

Based on the extended annotation guidelines, we manually annotated 12 plays for use as training data, chosen to broadly span the four hundred years of GerDraCor with respect to publication date, text length, and author gender (taken from GerDraCor metadata, Fischer et al. 2019, see Table 1). These 12 plays form the basis of the cross-validation experiments described in subsection 3.2, in which they are redistributed across folds rather than fixed into a single training and test split.

Author	Year	Title	Genre	In verse	Acts	Scenes	Speakers	SDs (Stage dirs.)	SDs within speech	SDs outside speech
Ayrer	1605	<i>Wie einem Weib jhr eygener Mann unwissenter Ding verkuppelt wird</i>	–	Yes	0	1	4	72	0	72
Borchert	1947	<i>Draußen vor der Tür</i>	–	No	0	7	16	135	34	101
Chézy	1823	<i>Der neue Narziß</i>	Lustspiel	Yes	0	7	11	52	8	44
Dohm	1878	<i>Ein Schuß in’s Schwarze</i>	Lustspiel	No	0	18	6	251	115	136
Dovsky	1915	<i>Mona Lisa</i>	–	Yes	2	0	20	0	0	0
Ebner-Eschenbach	1862	<i>Die Veilchen</i>	Lustspiel	No	1	3	8	169	68	101
Günderrode	1805	<i>Udohla</i>	–	Yes	2	0	7	41	0	41
Lessing	1772	<i>Emilia Galotti</i>	Trauerspiel	No	5	43	13	246	121	125
Neuber	1736	<i>Die von der Weisheit wider die Unwissenheit beschützte Schauspielkunst</i>	–	Yes	1	13	10	28	0	28
Sachs	1557	<i>Eulenspiegel mit dem blauen Hosentuch und dem Bauern</i>	–	Yes	0	1	4	22	0	22
Schiller	1781	<i>Die Räuber</i>	–	No	5	15	26	544	223	321
Wedekind	1891	<i>Frühlings Erwachen</i>	Tragödie	No	3	19	36	102	15	87

Table 1: Metadata of the 12 manually annotated plays. Genre was extracted from the self-ascriptions of a play, if available. A play counts as being written in verse when the number of <l> elements, i.e., lines, in the TEI source is higher than the number of <p> elements, i.e., paragraphs. Stage directions, abbreviated as SDs in the table, are the number of <stage> elements in the TEI source. Stage directions within a <l> or <p> element are counted as being within speech; all other stage directions are counted as being outside speech. The number of acts, scenes, and speakers was extracted via the DraCor API (<https://dracor.org/api/v1>).

3.2 Fine-tuning and Cross-validation Experiments

Based on the experiences from Guhr (2026), we used the TEI-enriching NEISS NTEE software developed by Lemke et al. (forthcoming) to fine-tune a pretrained German-language BERT model for the prediction and classification of sound events in GerDraCor. Originally developed for the task of named entity recognition, the software is specialized in automated TEI enrichment that goes beyond named entities and enables the recognition and classification of longer sequences rather than concentrating on single-word or token annotation. In other computational literary studies use cases, the software has been employed for the recognition and classification of spatial patterns (Zöllner et al. 2021), the detection of gender depiction (Flüh and Lemke 2022), and sound annotation in German- and English-language fiction (Guhr 2026; Guhr and Algee-Hewitt 2024; Guhr and Mahlberg 2026).

To assess how performance scales with the amount and type of training data, we conducted a series of 3-fold cross-validation experiments, progressively adding manually annotated prose texts from Guhr (2026) to the 12 manually annotated plays (see the workflow in Figure 2, and the distribution in Appendix A). In each fold, 4 of the 12 plays were held out as the test set and the remaining 8 used for fine-tuning, so that each play served as test data once and the results do not depend on a particular test set composition. The same 3-fold split was reused across all experiments to keep the configurations comparable. All scores reported below are macro-averages pooled over the twelve test plays, given as before and after postprocessing; detailed evaluation tables, including precision and recall, are provided in Appendix A.

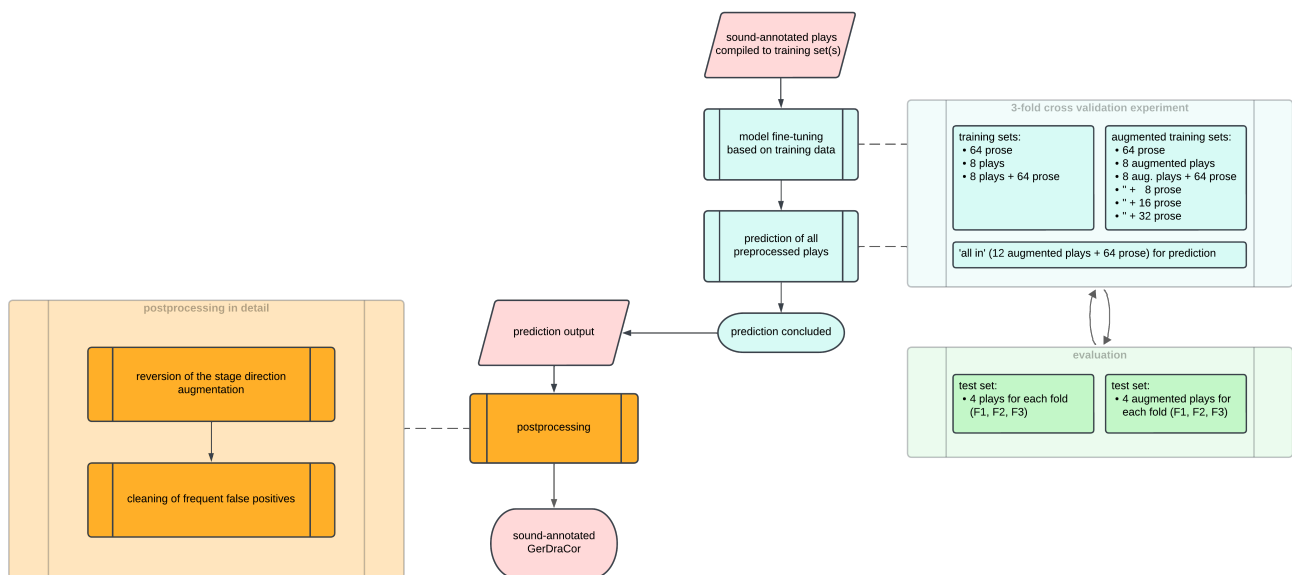


Figure 2: Workflow for automated sound annotation of GerDraCor plays, covering the fine-tuning 3-fold cross-validation experiments and their evaluation, as well as the postprocessing of the prediction output. Node colors indicate step type: orange = preprocessing and postprocessing steps; blue = manual annotation steps; green = evaluation; rose = data. The detail box on the left elaborates the preprocessing (stage direction augmentation via rule-based verb insertion). The detail boxes on the right elaborate the 3-fold cross-validation and evaluation steps.

Fold	Play	cs	as	Total
F1	Ayrer, <i>Fassnachtspiel. Wie einem Weib...</i>	88	0	88
	Borchert, <i>Draußen vor der Tür</i>	239	116	355
	Neuber, <i>Die von der Weisheit...</i>	22	12	34
	Wedekind, <i>Frühlings Erwachen</i>	97	12	109
	<i>F1 total</i>	446	140	586
F2	Chézy, <i>Der neue Narziß</i>	49	13	62
	Dohm, <i>Ein Schuß in's Schwarze</i>	170	7	177
	Dovsky, <i>Mona Lisa</i>	36	18	54
	Schiller, <i>Die Räuber</i>	447	102	549
	<i>F2 total</i>	702	140	842
F3	Ebner-Eschenbach, <i>Die Veilchen</i>	126	5	131
	Günderrode, <i>Udohla</i>	27	9	36
	Lessing, <i>Emilia Galotti</i>	195	7	202
	Sachs, <i>Eulenspiegel mit dem blauen...</i>	29	2	31
	<i>F3 total</i>	377	23	400
Total		1,525	303	1,828

Table 2: Gold sound event annotations per play in the augmented test set, grouped by cross-validation fold. cs = character sound events; as = ambient sound events.

All prose texts used for training are drawn from the second half of the 19th and the early 20th century (taken from the training set in Guhr (2026)), a period that partially overlaps with the publication period most densely represented in GerDraCor. This design allows a controlled assessment of the trade-off between the amount of training data and its genre specificity, that is, whether the gains from larger, prose-heavy training sets outweigh the domain mismatch they introduce.

The experiments proceed in four steps: (1) We first establish what a model trained exclusively on prose achieves on dramatic texts (prose only, 64 prose texts), then (2) what the manually annotated plays achieve on their own (Exp. 0), then (3) what their combination achieves (Exp. 1), and finally (4) whether assimilating the structure of the plays to that of prose via the heuristic changes the picture. Configurations trained on the augmented plays are marked with a prime: Exp. 0' and Exp. 1' repeat steps (2) and (3) on the augmented version, and Exp. 2' to Exp. 4' vary the amount of prose added to a constant set of eight augmented plays. All configurations are evaluated on the same twelve manually annotated plays. Because the augmentation modifies the text of the plays, the augmented and non-augmented conditions are evaluated against two versions of the test set, which contain 1,747 and 1,828 gold sound events, respectively; scores are therefore comparable within each version and not across them.

Postprocessing. Because the prediction output contains recurrent structural artifacts, all configurations are additionally evaluated after a postprocessing step. The script removes false positive annotations inside <speaker> elements, removes frequent false positives via list-matching, and merges duplicate annotations caused by spans interrupted by line breaks. The step is precision-oriented by design: It adds no new detections and can therefore only raise precision, convert partial matches into exact ones, and, occasionally, remove a prediction that did overlap a gold annotation. We report raw and postprocessed scores side by side throughout, since the size of the correction differs across configurations (see the Tables in Appendix A).

The Prose-only Baseline. As already mentioned briefly in [section 3](#), applying the model fine-tuned on the 64 sound-annotated prose texts of Guhr (2026) directly to the original plays yields a token macro F1 of 0.39 and an E-F1 of 0.35 before postprocessing. The model transfers a substantial part of what it has learned about sound events in prose to dramatic texts, but its errors are characteristic of a model that has never seen dramatic markup: It annotates speaker names inside <speaker> elements as character sounds, it misses sound events distributed across verse lines, and it under-predicts sound events in stage directions. The postprocessing step raises span detection precision from 0.58 to 0.71 and E-F1 from 0.35 to 0.40, but it cannot supply what the model does not predict: Span detection recall remains at 0.43, and 1,065 of the 1,747 gold sound events are missed, against 1,045 before postprocessing. The prose-only model is thus usable but not sufficient, and the deficit is concentrated in those features that distinguish drama from prose.

The Drama-only Baseline. For comparison, we fine-tuned a model on the manually annotated plays alone, holding out four plays per fold for evaluation as in all other configurations. The model reaches a token macro F1 of 0.24, well below the prose-only model, and ambient sound detection fails almost entirely (E-F1 = 0.01) because of the few ambient sound annotations in the training data. Accordingly, eight plays are too little training data for the task, and the contrast with the prose-only model shows that the quantity of genre-general sound event examples matters more than the genre fit of a small training set. This motivates the next step of combining the two training sets.

Combining Prose and Plays. Adding the 64 annotated prose texts to the eight non-augmented plays raises performance relative to the drama-only baseline (token macro F1 = 0.34 against 0.24), which confirms that prose data helps. However, relative to the prose-only model on the same original gold standard, adding the plays *lowers* performance, from 0.40 to 0.34 token macro F1, from 0.40 to 0.32 E-F1, and from 0.28 to 0.15 for ambient sound. Consequently, the manually annotated plays, added to prose in their original form, make the model worse rather than better.

The error profile shows that adding the plays slightly improves recall: False negatives fall from 1,045 to 1,013, most visibly in Ayres's *Fassnachtspiel* (38 to 24) and *Die Veilchen* (63 to 48). Precision and class discrimination deteriorate in exchange. False positives rise from 688 to 770, concentrated in the prose-dialogue plays *Draußen vor der Tür* (96 to 142) and *Emilia Galotti* (60 to 88), and label confusions rise from 75 to 92. The sound-annotated plays in the training data contain only a few examples of ambient sound, which biases the combined model toward the character sound class.

This highlights the differences between drama and prose and that a shared annotation scheme does work without any adaptation. The sound events of the prose training data are syntactically complete verbal phrases; a large part of the sound information in the plays is not. The distribution of the missed events across the corpus makes the point clear: The six plays written in verse hold 305 of the 1,828 gold events in the augmented test set but are missed at a consistently higher rate than the prose-dialogue plays in every configuration (see [subsection 3.3](#)). Elliptical stage directions are a substantial part of the problem, though not the largest. We therefore examined whether model performance benefits from an assimilation of the plays' structure to prose.

Configuration	E-F1	E-F1 [cs]	E-F1 [as]
Prose only, on augmented plays	0.32 0.37	0.34 0.40	0.11 0.14
Exp. 1' (8 drama aug. + 64 prose)	0.41 0.46	0.44 0.49	0.16 0.19
Exp. 3' (8 drama aug. + 16 prose)	0.43 0.47	0.47 0.51	0.16 0.19

Table 3: E-F1 by sound class on the augmented test set (gold = 1,828), before | after postprocessing.

Assimilating the Plays to Prose. As a consequence of the preceding result, we recognized that the challenge in sound prediction is of a structural rather than a semantic nature. This motivated our heuristic approach, in which we investigated whether adding plays would improve performance if their sound indications resembled the prose training data, and repeated the combination experiment.

Surprisingly, the result reverses the sign of the drama contribution. In their original form, the plays cost six points of token macro F1 when added to prose; in augmented form they gain between three and four points. Against a prose-only model evaluated on the same augmented plays and the same gold standard, adding the eight augmented plays improves every relevant metric after postprocessing: E-F1 rises from 0.37 to 0.46 in Exp. 1' and to 0.47 in Exp. 3', Gamma from 0.51 to 0.57, and span detection F1 from 0.58 to 0.62.

The detailed results in Appendix A identify what the drama data actually contributes to the fine-tuning. Relative to the prose-only model on the same gold standard, span detection recall gains six points (0.57 against 0.51 in Exp. 1', 0.58 in Exp. 3') at essentially unchanged precision (0.73 and 0.70 against 0.71), and E-F1 rises by nine and eleven points, while token macro F1 rises by only three and four. Accordingly, adding the plays contributes comparatively little to what the model recognizes as a sound event, which the prose data has already established. Their contribution is concentrated in where a sound event begins and ends in a dramatic text, which is the division of labor between genre-specific structural learning and genre-general sound event recognition that we hypothesized in section 1.

The gain, however, is not evenly distributed across the two sound classes (Table 3). Adding the augmented plays raises ambient E-F1 from 0.14 to 0.19, but the character sound class gains roughly twice as much, from 0.40 to 0.49 and 0.51, so that the distance between the two classes widens rather than closes. Ambient sound detection remains the weakest part of the output in every configuration, at little more than a third of the character sound level, which is unsurprising given how few ambient sound annotations the dramatic training data contains.

Performance also varies considerably across folds, reflecting the structural diversity of the corpus discussed in subsection 3.1: In Exp. 1', per-play Gamma after postprocessing ranges from 0.40 to 0.76, and the fold containing *Die Räuber* and *Der neue Narziß* is the weakest in every configuration we evaluated. Consequently, the variation is partly a property of the test sets rather than of the models, because the four plays held out in fold 1 contain 140 ambient sound events, while those held out in fold 3 contain 23. We therefore report scores pooled across all twelve test plays throughout and do not select individual folds.

Experiment	Token Macro F1	Span Det. F1	E-F1	Gamma
<i>Original plays (gold = 1,747)</i>				
Prose only (64 prose)	0.39 0.40	0.48 0.52	0.35 0.40	0.43 0.48
Exp. 0 (8 drama original)	0.24 0.24	0.47 0.55	0.30 0.32	0.45 0.47
Exp. 1 (8 drama orig. + 64 prose)	0.33 0.34	0.48 0.51	0.27 0.32	0.40 0.45
<i>Augmented plays (gold = 1,828)</i>				
Prose only, on augmented plays	0.37 0.38	0.53 0.58	0.32 0.37	0.45 0.51
Exp. 0' (8 drama aug., +0 prose)	0.33 0.34	0.56 0.58	0.38 0.39	0.48 0.51
Exp. 1' (8 drama aug. + 64 prose)	0.40 0.41	0.59 0.62	0.41 0.46	0.52 0.57
Exp. 2' (8 drama aug. + 8 prose)	0.37 0.37	0.59 0.60	0.39 0.43	0.51 0.54
Exp. 3' (8 drama aug. + 16 prose)	0.41 0.42	0.59 0.62	0.43 0.47	0.53 0.57
Exp. 4' (8 drama aug. + 32 prose)	0.44 0.45	0.57 0.60	0.39 0.44	0.51 0.55

Table 4: Evaluation scores pooled over all twelve test plays, reported as before | after postprocessing with best values per metric in bold.

Finding the Right Training Set Distribution. The remaining question is how much prose the augmented plays need alongside them. Holding the eight augmented plays constant, we varied the amount of added prose across four configurations: 8 texts (Exp. 2'), 16 (Exp. 3'), 32 (Exp. 4'), and 64 (Exp. 1'), and recognized that model performance does not increase monotonically with the amount of prose. Exp. 3', with 16 prose texts, is the best configuration on span detection F1, E-F1, and Gamma, while the largest prose addition in Exp. 1' performs slightly below it on all three; only token macro F1 peaks later, in Exp. 4'. This means that beyond sixteen prose texts, further prose weakens the drama-specific structural signal more than it adds usable sound event examples.

The Effect of Postprocessing. In Exp. 1', the step reduces the number of predicted spans from 1,687 to 1,386 and removes 41% of all false positives (676 to 397), while false negatives rise only marginally (817 to 839). Span detection precision increases from 0.63 to 0.73 at a recall cost of less than one point. The gains are largest for the strictest criteria, which follows from how the step operates: Removing structural artifacts and merging fragmented spans adds no new detections but converts partial matches into exact ones. E-F1 rises from 0.41 to 0.46, span detection under IoU ≥ 0.5 from 0.50 to 0.55, and Gamma from 0.52 to 0.57. Merging also reduces label confusions, since differently labeled fragments of one gold span count as confusions before merging; their number falls from 79 to 61.

However, the size of the correction depends on the configuration rather than being constant across them. While the postprocessing step removes 51% of the false positives produced by the prose-only model on the augmented plays and 44% of those it produces on the original plays, it removes only 41% in Exp. 1' and 37% in Exp. 3'. Interestingly, this ordering does not follow the raw false positive count: Exp. 3' produces almost as many false positives as the prose-only model (856 against 884), but benefits least from their removal. The reason is that its surplus predictions are speech-describing expressions, while the prose-only model mainly over-predicts speaker names as character sounds (see subsection 3.3). Accordingly, the more prose-heavy the training data, the more of the model's errors are structural artifacts that get removed by the postprocessing.

At the same time, the comparison delineates what postprocessing cannot do. Token-level scores move only slightly, from 0.404 to 0.414 token macro F1 in Exp. 1', because the step repairs the span structure of the output without altering the classifier decisions, and the

qualitative false positive types identified in [subsection 3.3](#) persist. Postprocessing also cannot create recall: Ambient sound detection remains the weakest part of the output, at an E-F1 of 0.16 before and 0.19 after the step.

3.3 Qualitative Error Analysis

We conducted a qualitative error analysis on the pooled predictions for all twelve test plays across the three cross-validation folds. False positives and false negatives were determined on the raw prediction output, using any-overlap matching with label agreement, so that a correctly located span carrying the wrong label counts as both. The aggregate counts in [Table 5](#) locate the contribution of the drama training data on the recall side.

False positives. Four error types recur across all three configurations.

1. Physical action verbs in stage directions are annotated as sound events, for instance “aufspringt” (‘jumps up’) in *Emilia Galotti*, “stürzt zum kleinen Tische” (‘rushes to the small table’) in *Die Veilchen*, or “zupft ihn” (‘tugs at him’) in *Eulenspiegel*, actions that may incidentally produce sound but do not constitute sound events in the annotation scheme.
2. Speech-describing expressions in dialogue that do not represent a realized sound event are over-predicted, such as “Und ich soll Ihnen noch lange erzählen” (‘And I am supposed to keep telling you for a long time’) in *Emilia Galotti* or “sagt die Schrift” (‘the scripture says’) in *Die Räuber*, indicating that the boundary between a realized sound event and a nonevent remains a persistent challenge (Gius and Vauth 2022).
3. Over-segmentation splits single continuous sound events into fragments, which the span-based evaluation counts as both false positives and false negatives.
4. A fourth type is structural rather than semantic: Annotations of speaker names inside <speaker> elements. It is most pronounced in the prose-only condition, which has encountered no dramatic markup in training and has no basis for treating a speaker prefix differently from narrative text, but it is not confined to it. The postprocessing removes this type entirely.

False negatives. The false negative rate reflects three structural challenges.

1. Verse-internal sound events are disproportionately missed, and the gap widens as drama training data is added. Of the 1,828 gold events, 305 occur in the six plays written in verse. The prose-only model misses 50.2% of them against 47.9% in the prose-dialogue-heavy plays; in Exp. 1’, the percentage decreases by six points (49.5 against 43.7%), and in Exp. 3’ by close to eight (47.2 against 39.5%). Misses such as “Die zarten Halmen flüstern es sich zu” (‘The delicate blades of grass whisper it to each other’) in Günderröde’s *Udohla* occur in all three configurations, and the augmentation heuristic, which operates on stage directions, does not address them, which is mainly due to too few examples of the double-annotation decision for sound events separated by line breaks.

Configuration	<i>before postpr.</i>			<i>after postpr.</i>		
	Spans	FP	FN	Spans	FP	FN
Prose only (64 prose)	1,829	884	883	1,336	430	922
Exp. 1' (8 drama aug. + 64 prose)	1,687	676	817	1,386	397	839
Exp. 3' (8 drama aug. + 16 prose)	1,939	856	745	1,601	543	770

Table 5: Predicted spans, false positives, and false negatives pooled over all twelve test plays, scored against the augmented gold standard of 1,828 sound events. Any-overlap matching with label agreement.

2. Elliptical stage directions are frequently missed. This is the gap the augmentation heuristic targets, but some false negatives remain stable across configurations: Missed by the prose-only model, by Exp. 1', and by Exp. 3' alike are, e.g., "meldend" ('announcing') in *Die Veilchen* and "Für sich" ('To himself') and "laut" ('loudly') in *Ein Schuß in's Schwarze*. Elliptical stage directions are therefore an important source, but not the principal one, of missed sound events.
3. Onomatopoetic sound representations are missed almost entirely. Borchert's *Draußen vor der Tür* renders knocking as the repeated exclamation "Rums", which no configuration detects in any of its occurrences, and the play's soundscape-setting stage directions such as "Die Elbe schwapppt gegen die Pontons" ('The Elbe laps against the pontoons') are likewise missed throughout. *Draußen vor der Tür* accordingly remains the play with the largest absolute number of missed events after *Die Räuber* in every configuration.

Classification errors. Confusions between the two sound classes are rare relative to detection errors, and concentrated in a small part of the corpus: *Draußen vor der Tür* and *Die Räuber*, the two plays with by far the most ambient annotations in the test set, account for ca. two thirds of the confusions in every configuration. The model's difficulty is thus specific to the ambient/character boundary, which is only at issue where ambient sound is present, rather than a general weakness in distinguishing the two classes. A recurring trigger is the generic-speaker construction with "man" or an unspecified plural subject, as in "Man sagt" ('One says') or "und sprach" ('and spoke'): The *verba dicendi* signal speech, which the model reads as character sound, whereas the annotation scheme classifies them as ambient because no identifiable character can be determined as the source, which is a conceptual rather than a lexical difficulty.

Borderline predictions. A further category concerns predictions that are plausible from an auditory perspective but do not satisfy the annotation scheme's requirement of an explicit sound word. In *Draußen vor der Tür* the model annotates the stage direction "Man hört den Wind und das Wasser" ('One hears the wind and the water') as an ambient sound in both combined configurations. The gold standard leaves the passage unannotated because "Wind" ('wind') and "Wasser" ('water') are not sound words and the auditory information is carried solely by the perception verb "hören". Such predictions are counted as false positives although they arguably capture information about the fictional soundscape, pointing to a limitation of the explicit sound word-based operationalization when applied to perception-verb constructions.

Choice of the Deployed Model. For the corpus-wide prediction, we fine-tuned a model on all twelve annotated plays together with the 64 prose texts. Because no play remains held out, this model cannot be evaluated directly; we take the scores of Exp. 1' as its performance estimate, since its training data differs only in that four of the twelve plays are withheld in each fold.

While Exp. 1' is not the strongest configuration on the aggregate metrics, and Exp. 3' outperforms on token macro F1 (0.42 against 0.41) and E-F1 (0.47 against 0.46), and marginally on span detection F1 and Gamma (see Table 4), we nevertheless based the corpus-wide prediction on the Exp. 1' configuration because its error profile is better suited to a corpus-level analysis. As Table 5 shows, it produces the fewest false positives of the configurations we compared, 397 after postprocessing against 543 for Exp. 3', at the cost of 69 additional false negatives. For the density-based analyses in section 4, false positives distort the measure more than missed events, since they add sound events that are not in the text. Regarding ambient sound, both configurations perform equivalently poorly.

3.4 Sound Event Density Calculation

To capture how frequently sounds are represented in relation to a play's overall length and to get comparable data to analyze potential changes in sound representation in drama over time, we calculate the sound event density (SE_D) for each annotated GerDraCor play using the SE_D measurement developed by Guhr (2026).

$$SE_D = \frac{T_{se}}{T_{pe}} \times 100$$

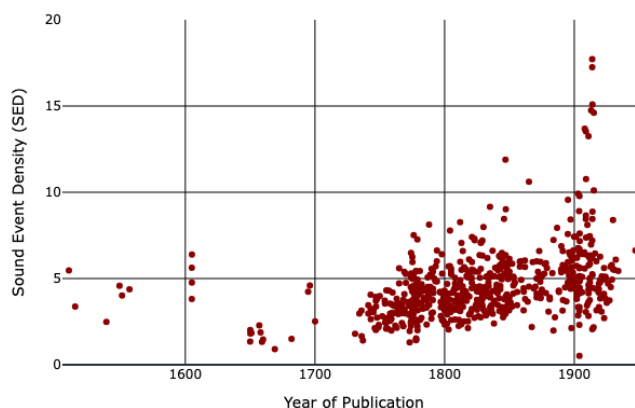
The measurement relies on the absolute count of annotated sound events (T_{se}) and the calculation of the estimated count of potential events in a text (T_{pe}). The latter is calculated as the average span length (in words) of all annotated sound events in the text and dividing the total word count by this value, yielding an approximate number of possible events. The actual number of annotated sound events is then divided by this estimate and multiplied by 100, producing a density value that can be compared across plays of different lengths, genres, or historical periods.

4. Sound Analysis of GerDraCor

For the analysis of explicit sound events in GerDraCor, we predicted the augmented version of the plays with the 'all in' model in the NEISS N-TEE environment³, applied the postprocessing script, and reversed the *sagt* and *sagen* inversion. Since the play augmentation step also alters whitespace and line breaks in the TEI source in ways a regular expression cannot reliably distinguish from the original encoding, the annotated corpus we analyze and release retains these changes and is not byte-identical to the GerDraCor source.

3. For reasons we could not identify, the model only returned sound annotations for 650 of the 769 plays in the corpus; the plays with empty output are excluded from the analysis.

Sound Event Density per Text by Year (All Genres, n=650)

**Figure 3:** SE_D of 650 GerDraCor plays.

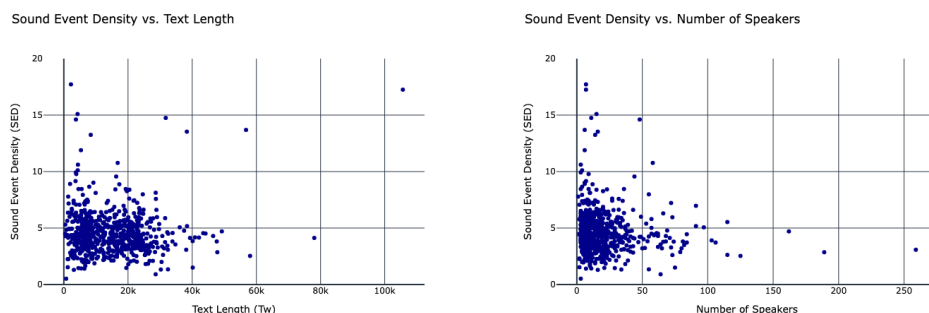
Before presenting the analysis results, a note on reliability is warranted: As the cross-validation experiments demonstrate, the model does not achieve perfect span-level accuracy, and the results below should be interpreted with this margin of error in mind. None of this, however, compromises comparison within the corpus. Since the same model was applied uniformly, any systematic errors affect all plays alike rather than biasing particular periods, genres, or authors, and the model under-predicts consistently rather than selectively (see [subsection 3.3](#)). The analyses that follow are therefore best understood as comparative rather than absolute: The predicted SE_D values do not measure the true density of sound events, but they provide a consistent and reproducible basis for identifying patterns and differences across the corpus. We first trace sound event density across the four centuries the corpus spans ([subsection 4.2](#)), and then compare plays grouped by literary movement and by author ([subsection 4.3](#)).

4.1 Sound Event Density of GerDraCor

For the diachronic analysis of GerDraCor, we visualized the SE_D values against normalized publication dates from DraCor metadata (see [Figure 3](#)). The mean sound event length of all sound-predicted plays is 2.9⁴.

A correlation analysis between text length and SE_D (see [Figure 4a](#)) reveals a very weak positive relationship ($r = 0.068$, $p = 0.0971$), which falls well short of statistical significance at the conventional 0.05 threshold. In practical terms, the density of sound events appears largely independent of dramatic text length, implying that the SE_D metric effectively normalizes for overall textual size, with only negligible residual influence. Another correlation analysis, this time between the number of speakers and SE_D (see [Figure 4b](#)), reveals a very weak negative relationship ($r = -0.062$, $p = 0.1147$) that does not reach statistical significance, suggesting that SE_D is largely independent of the number of speakers in a text. Plays by Ludwig Thoma again rank among the texts with high SE_D , consistent with earlier findings on his prose works (Guhr 2026). However, within the dramatic corpus, a new author emerges as particularly prominent in terms

4. In prose texts, this number is 4.39 (Guhr 2026).



(a) Text length and SE_D , with the outlier play *Ignorabimus* by Holz in the upper right corner. The three plays in the upper left corner are by Stramm, the three in the upper center by Bahr and Holz. (b) Number of speakers and SE_D , with the outlier play *Napoleon oder Die hundert Tage* by Grabbe in the lower right corner. Holz's *Ignorabimus* and Stramm's *Sancta Susanna* are in the upper left corner.

Figure 4: Correlation between SE_D and two play-level measures: text length (a) and number of speakers (b).

of SE_D : August Stramm. Of his six plays, four exhibit SE_D values above 8. Similarly, Hermann Bahr stands out, with both of his plays showing SE_D scores exceeding 10. The works of Stramm and Bahr are also particularly noticeable in the visualization of the relationship between SE_D and text length, as are the plays by Arno Holz (see Figure 4a)⁵.

4.2 Sound Event Density across Four Centuries

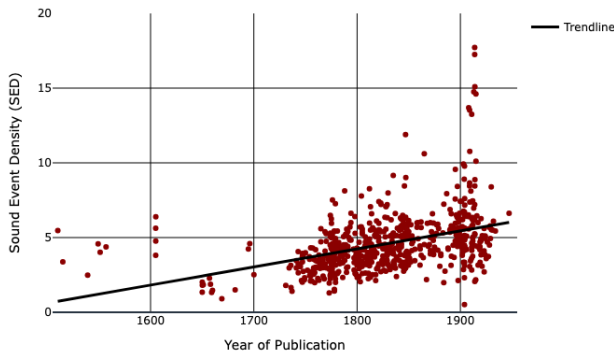
To investigate how explicit sound indications develop across the four centuries covered by GerDraCor, we analyze the SE_D of the predicted plays in relation to their publication dates. The upward trend in Figure 5a raises an interpretive question: Does the increase reflect a genuine enrichment of dramatic soundscapes, or does it mirror the historical expansion of stage directions as a dramaturgical convention (Trilcke et al. 2020)? The trend itself is statistically robust but weak (Spearman's $\rho = 0.52$, $p < 0.001$), so year alone is not a sufficient explanation.⁶ That's why stage direction density (stage direction words per 1,000 spoken words) is the stronger predictor: Across 601 plays, it correlates with SE_D at $\rho = 0.68$ ($p < 0.001$), and the relationship holds after controlling for the temporal trend the two share (partial $\rho = 0.55$, $p < 0.001$), while text length confounds neither measure ($p > 0.25$). Period medians for stage direction density rise monotonically from 27 words per 1,000 in the earliest period to 170 post-1900 (Figure 6). The SE_D increase is therefore at least partly a consequence of changing dramaturgical convention rather than exclusively of a shift toward more explicit sound in younger plays. The two endpoints of our annotated set illustrate the same divergence: Ayres's *Fassnachtspiel* (1605) is unusually direction-heavy for its period, Borchert's *Draußen vor der Tür* (1947) unusually sparse, so neither tracks the corpus-level trend, and the corpus analysis rather than individual plays is the basis for our diachronic claims.

5. Interactive versions of all figures with text titles as hover elements can be found in the archived GitHub Repository.

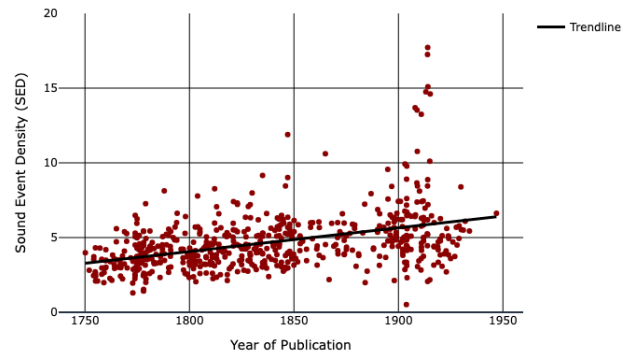
6. A linear regression confirms the trend ($\beta = 0.016$, $p < 0.0001$, $R^2 = 0.175$), but the residuals show strong skew and kurtosis, so Spearman's ρ is our primary statistic throughout.

The effect, however, is specific to one sound class: Stage direction density correlates with character SE_D at $\rho = 0.67$ but with ambient SE_D at only $\rho = 0.38$, so what grows over time is the explicit qualification of speech acts rather than the description of environmental soundscapes. The Spearman correlation matrix in Figure 9 confirms this: Stage direction density, year, and the SE_D measures form a coherent cluster, while text length and number of speakers, though strongly correlated with each other ($\rho = 0.63$), show no meaningful relationship with sound event density.

Sound Event Density per Text by Year (All Genres, n=650)

(a) All sound-predicted GerDraCor texts, with a trend line showing the increase in SE_D over the period.

Sound Event Density per Text by Year (1750-1950, n=599)



(b) The same data filtered for plays published between 1750 and 1950.

Figure 5: SE_D across GerDraCor: The predicted corpus with trend line (a) and the subset with plays published between 1750 and 1950 (b). Starting from the top (decreasing from $SE_D = 17.72$ to $SE_D = 13.24$), the outliers on the upper right are Stramm's *Sancta Susanna*, Holz's *Ignorabimus*, Stramm's *Erwachen*, Bahr's *Das Phantom*, Stramm's *Geschehen*, Holz's *Sonnenfinsternis*, Bahr's *Das Konzert*, and Sorge's *Odysseus*.

SED and stage direction density by half-century period

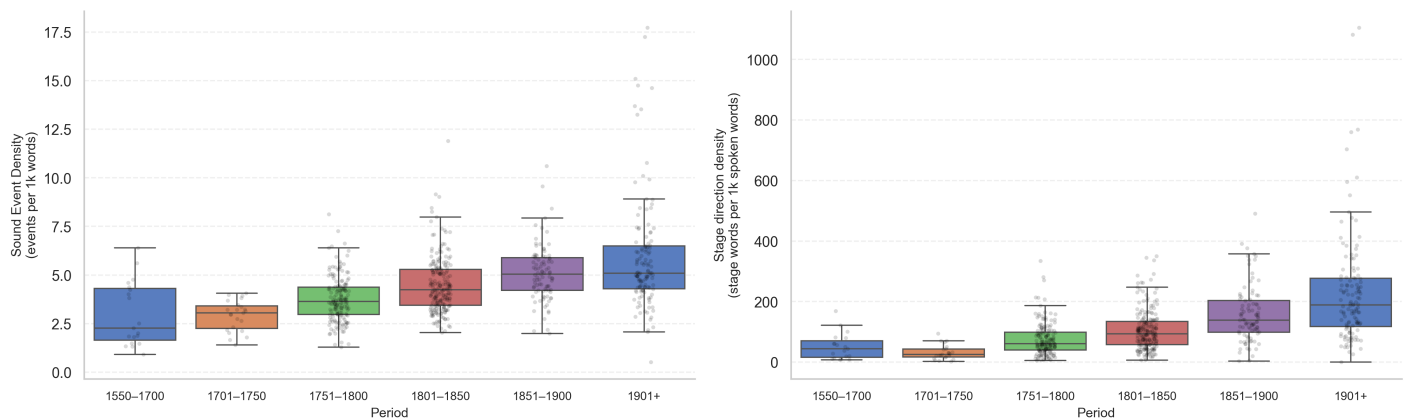


Figure 6: SE_D (left) and stage direction density (right) by half-century period across 650 GerDraCor plays. Both measures increase monotonically from the earliest period to the post-1900 period, with the most pronounced growth visible from 1801–1850 onward. The wider spread and higher outliers in the later periods reflect the growing stylistic diversity of the corpus, particularly the influence of Naturalist and Expressionist authors.

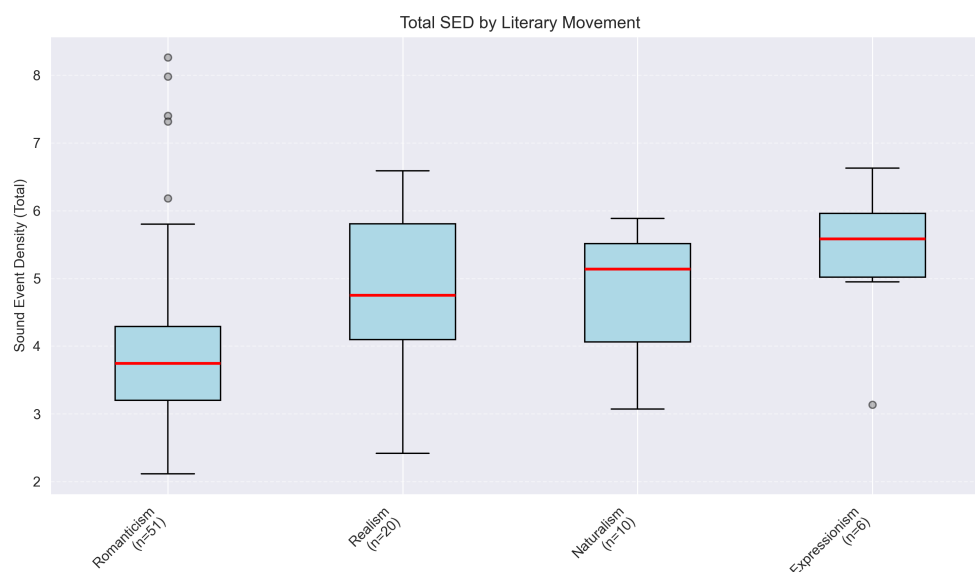


Figure 7: SE_D of selected literary movements.

4.3 Sound Event Density across Literary Movements

To investigate whether the SE_D varies across literary movements, we filtered the 650 sound-annotated plays for plays published between 1750 and 1950 by author and literary movement (see Figure 5b).

Given that most GerDraCor plays have a publication date between 1750 and 1950 ($n=678$ in the entire corpus; $n=599$ in our sound-annotated subset of 650 plays), we concentrated our SE_D analysis on this period (see Figure 5b) and could recognize a statistically significant upward trend (slope = 0.017, $p < 0.0001$) with a particularly high proportion of high- SE_D plays in the first quarter of the 20th century. In an additional enrichment of the metadata, we filtered the plays by author name and their relation to the chosen literary movements and visualized the SE_D values as box plots in Figure 7. For the choice of authors, we consulted the list of authors of the corpora *d-RoRo* (Schumacher et al. 2022), and the literary movement subcorpora of *d-Prose* (Gius et al. 2021)⁷.

Among these literary movements, we observed highly significant differences overall⁸ and subsequent pairwise post-hoc Dunn's tests revealed significant differences between Romanticism and Realism ($p \approx 0.0006$), extremely significant differences between Romanticism and Naturalism ($p \approx 2.8 \times 10^{-12}$), significant differences between Romanticism and Expressionism ($p \approx 7 \times 10^{-6}$), and between Realism and Naturalism ($p \approx 0.0045$). Differences between Realism and Expressionism, as well as between Naturalism and Expressionism, were not statistically significant.

However, since the classification of literary movements is based solely on authors' names and ignores how some authors' writing has developed – only think about the literary

7. Romanticism authors: Arnim, Brentano, Eichendorff, Fouqué, Heine, Günderrode, Kleist, Klingemann, Moritz, Schiller, Schlegel, Tieck, Uhland, Zschokke; Realism authors: Klemm, Koch, Laube, Reuter, Scheerbarth, Ebner-Eschenbach, Wildenbruch; Naturalism authors: Anzengruber, Bahr, Bierbaum, Bleibtreu, Dohm, Ganghofer, Hauptmann, Holz, Sudermann; Heym, Lasker-Schüler, Heym, Kaiser, Kafka, Toller, Sternheim, Stramm, Wedekind.

8. $p < 0.0001$; The Kruskal–Wallis test was chosen for significance testing as it compares more than two groups without assuming normality.

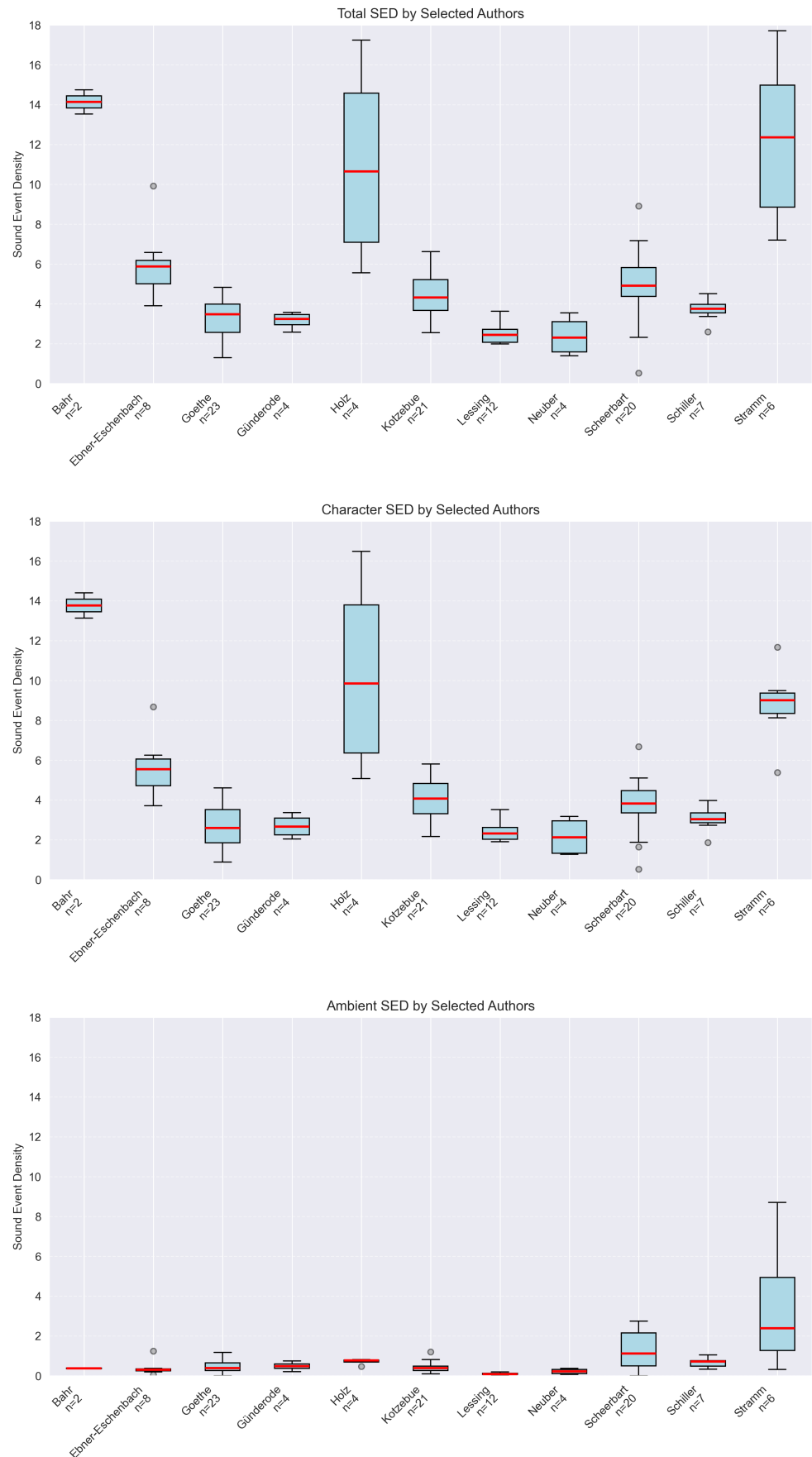


Figure 8: Plots of total SE_D of plays filtered by author.

development of Goethe's writing – the comparison of the SE_D of literary movements must be treated with caution. Accordingly, we visualized the SE_D scores of some authors from this period (see Figure 8), regardless of publication date or literary movement. The authors were chosen based on the number of their plays with sound annotations in GerDraCor, specifically Goethe (23), Kotzebue (21), and Scheerbart (19). We also included Schiller (8) and Lessing (12), as well as the three SE_D outlier authors, Bahr (2), Holz (2), and Stramm (6); and the three female authors with the most plays in GerDraCor: Ebner-Eschenbach (8), Neuber (4), and Günderrode (4). Looking at the box plots, we can again recognize the outlier authors Holz, Bahr, and especially Stramm, who have high overall SE_D scores for all their plays combined. Ebner-Eschenbach's plays also show an elevated SE_D . When comparing them to the other authors in the plot, however, the low number of plays has to be taken into account: Holz and Bahr are represented by two plays each, and the two female authors Neuber and Günderrode by four plays each.

Example 7. <sp who="laszlo"> <speaker>Laszlo.</speaker>

<p>... jagt ... <ambient_sound>und ratter</ambient_sound> ... <ambient_sound>und knatter</ambient_sound> und wirbelt ... <stage><character_sound>Lacht auf</character_sound> und wirft die Pfeife in die Ecke.</stage></p> </sp>

<sp who="laszlo"> <speaker>Laszlo.</speaker>

<p>... chases ... <ambient_sound>and clatters</ambient_sound> ... <ambient_sound>and rattles</ambient_sound> and swirls ... <stage><character_sound>Laughs</character_sound> and throws the pipe into the corner.</stage></p> </sp> (*Die Haidebraut*, Stramm)

Among the outliers in the ambient SE_D scores, Stramm's plays are especially well represented. For instance, the expressionist's rather short play, *Die Haidebraut* (3,063 words), provides information about the rich soundscape of the dramatic world, referring to rattling and laughing, as seen in Example 7. Other sounds in Stramm's play include screaming, shouting, knocking, and the sound of a collapsing hut ("zusammenknatternde[] Hütte") or the rustling of the forest. However, with a proportion of 0.3 ambient sounds among the total number of sound events, Stramm's plays do not have a particularly higher proportion compared to other authors' texts. For comparison, Goethe's plays have a proportion of 0.23 ambient sound events among the total number of sound events.

Taken together, the diachronic, movement-level, and author-level results describe the same corpus from three angles: Sound representation in German drama grows over four centuries, partitions by literary movement, and varies enough between authors to cut across those groupings.

5. Conclusion and Outlook

Adapting sound recognition from prose to drama proved to be a structural rather than a semantic problem. Both genres draw on the same sound vocabulary, but drama distributes it differently: Lacking a narrator, plays split auditory information between the primary text of dramatic speech and the secondary text of stage directions. In addition, a substantial part of it is elliptical, carried by manner adverbials that indicate

how an utterance sounds without naming the speech act. These are precisely the features that a model trained on prose cannot process.

Our cross-validation experiments confirm H1.1 and qualify H1.2: A model fine-tuned on prose alone transfers only partially to plays (token macro $F1 = 0.39$), and a model fine-tuned on the annotated plays alone performs worse still (0.24), so neither source suffices on its own. Combining them raises performance only once the plays have been assimilated to the syntax of prose: Added in their original form, they cost six points of token macro $F1$, whereas in augmented form they gain three to four. Accordingly, our augmentation heuristic prepares the plays for easier prediction with a model fine-tuned on prose-heavy data, meaning that the division of labor we hypothesized in [section 1](#) holds, although less cleanly than expected: The contribution of the plays is concentrated in span structure, where E- $F1$ rises by nine to eleven points, and is smaller at the token level. Furthermore, it is unevenly distributed across the two sound classes, since ambient sound remains the weaker one in every configuration, which follows from how rarely the dramatic training data marks it.

With regard to RQ2, sound representation increases significantly across the four centuries of the corpus (H2.1, Spearman's $\rho = 0.52$), most markedly in the first quarter of the 20th century. Its correlation with stage direction density ($\rho = 0.68$) indicates that this trend is at least partly mediated by the historical expansion of stage directions as a dramaturgical convention rather than solely by a shift toward more sound-conscious aesthetics. The effect is moreover specific to character sound, which correlates with stage direction density at $\rho = 0.67$ against only $\rho = 0.38$ for ambient sound. Sound indication also differs significantly across literary movements (H2.2), most sharply between Romanticism on the one hand and the Naturalist and Expressionist plays on the other, while individual authors, in particular Stramm, Bahr, and Holz, stand out from the movements they are assigned to.

The principal limitation of our approach is the size of the annotated drama set: Eight plays per fold are too few, and prose data compensates only in part. Since 64 manually annotated prose texts yielded $F1$ scores above 0.7 for the same task in Guhr (2026), future work should consider annotating more plays to better performance. This supports Weitin (2021, 86), who emphasizes that the training data ultimately determine the output of any classification task. Within that constraint, however, the approach is productive: The model detects a substantial share of the explicit sound indications in 650 plays, and it does so uniformly enough that the resulting SE_D values support comparison across periods, movements, and authors, even where individual spans are missed. Future work could extend the annotation to loudness levels, establish inter-annotator agreement for the drama-specific extensions to the guidelines, and relate the textual sound profiles to documented performance practices.

6. Data Availability

Data can be found here: <https://github.com/dracor-org/gerdracor/tree/main/tei>

7. Software Availability

Software can be found here: https://github.com/SvenjaGuhr/Sound_in_Drama

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9. Author Contributions

Svenja Guhr: Project administration, Conceptualization, Data curation, Methodology, Formal analysis, Writing – original draft

Janis Pagel: Conceptualization, Data curation, Methodology, Writing – review & editing

Nils Reiter: Conceptualization, Methodology

References

- Agapitos, Paschalis and Andreas van Cranenburgh (2024). “A Stylometric Analysis of Seneca’s Disputed Plays. Authorship Verification of *Octavia* and *Hercules Oetaeus*”. In: *Journal of Computational Literary Studies*. 10.48694/JCLS.3919.
- Aston, Elaine and George Savona (1991). *Theatre as Sign-system : A Semiotics of Text and Performance*. Routledge. <http://lib.leeds.ac.uk/record=b1177901> (visited on 05/09/2025).
- Chan, Branden, Stefan Schweter, and Timo Möller (2020). “German’s Next Language Model”. In: *Proceedings of the 28th International Conference on Computational Linguistics*. International Committee on Computational Linguistics, 6788–6796. 10.18653/v1/2020.coling-main.598.
- DeepL SE (2026). *DeepL Translator*. <https://www.deepl.com/translator> (visited on 05/10/2026).
- Dennerlein, Katrin, Thomas Schmidt, and Christian Wolff (2024). “Emotions in Stage Directions in German Drama of the Early Modern Period: Explorations via Computational Emotion Classification”. In: *Computational Drama Analysis*. Ed. by Melanie Andresen and Nils Reiter. De Gruyter, 189–216. 10.1515/9783111071824-010.
- Fischer, Frank, Ingo Börner, Mathias Göbel, Angelika Hechtel, Christopher Kittel, Carsten Milling, and Peer Trilcke (2019). “Programmable Corpora: Introducing DraCor, an Infrastructure for the Research on European Drama”. In: *Proceedings of DH2019: “Complexities”*. Utrecht, The Netherlands. 10.5281/zenodo.4284002.

- Fischer, Frank, Peer Trilcke, Christopher Kittel, Carsten Milling, and Daniil Skorinkin (2018). "To Catch a Protagonist: Quantitative Dominance Relations in German-language Drama (1730–1930)". In: *Book of Abstracts of DH2018*. Mexico: ADHO, 193–201. <https://dh2018.adho.org/to-catch-a-protagonist-quantitative-dominance-relations-in-german-language-drama-1730-1930/> (visited on 12/05/2019).
- Flüh, Marie and Marc Lemke (2022). "An Experimental Attempt to Use Transfer Learning for Named Entity Recognition in Letters from the 19th and 20th Century". In: *DH 2022. Book of Abstracts*. <https://dh2022.dhii.asia/dh2022bookofabsts.pdf> (visited on 12/28/2022).
- Foley, Matt (2023). *Gothic Voices: The Vococentric Soundworld of Gothic Writing*. 1st ed. Cambridge University Press. [10.1017/9781009162579](https://doi.org/10.1017/9781009162579).
- Gius, Evelyn, Svenja Guhr, and Benedikt Adelmann (2021). *d-Prose 1870-1920*. Zenodo. [10.5281/zenodo.5015008](https://doi.org/10.5281/zenodo.5015008).
- Gius, Evelyn and Michael Vauth (2022). "Towards an Event Based Plot Model. A Computational Narratology Approach". In: *Journal of Computational Literary Studies* 1 (1). [10.48694/jcls.110](https://doi.org/10.48694/jcls.110).
- Glotova, Elena (2021). "Soundscapes in Nineteenth-century Gothic Short Stories". Doctoral Thesis. Umeå University. <https://urn.kb.se/resolve?urn=urn:nbn:se:umu:diva-183080> (visited on 01/27/2024).
- Guhr, Svenja (2026). *Raise Your Voice. Character Sound in German-language Fiction*. Metzler Series: Digitale Literaturwissenschaft. in print. Springer/Nature.
- Guhr, Svenja and Mark Algee-Hewitt (2024). "What's that Scary Sound? Ambient Sound in Gothic Fiction". In: *Journal of Computational Literary Studies* 2 (1). [10.48694/jcls.3583](https://doi.org/10.48694/jcls.3583).
- Guhr, Svenja and Michaela Mahlberg (2026). "Soundful Dickens. The Narrative Functions of Sounds in Dickens's Fictional Worlds". In: *Book of Abstracts of DHd2026*. Ed. by Silke Schwandt, Gabriel Viehhauser, Tara Andrews, Thomas Wallnig, Beatrice Nava, and Patrick Helling. Zenodo. [10.5281/ZENODO.18696290](https://doi.org/10.5281/ZENODO.18696290).
- Hillebrandt, Claudia (2018). "Literaturwissenschaft". In: *Handbuch Sound: Geschichte – Begriffe – Ansätze*. Ed. by Daniel Morat and Hansjakob Ziemer. J.B. Metzler, 120–125. [10.1007/978-3-476-05421-0_22](https://doi.org/10.1007/978-3-476-05421-0_22).
- Karsdorp, Folgert, Mike Kestemont, Christof Schöch, and Antal van den Bosch (2015). "The Love Equation: Computational Modeling of Romantic Relationships in French Classical Drama". In: *6th Workshop on Computational Models of Narrative (CMN 2015)*. Ed. by Mark A. Finlayson, Ben Miller, Antonio Lieto, and Remi Ronfard. OpenAccess Series in Informatics (OASIs) 45. Schloss Dagstuhl–Leibniz-Zentrum fuer Informatik, 98–107. [10.4230/OASIs.CMN.2015.98](https://doi.org/10.4230/OASIs.CMN.2015.98).
- Kendrick, Lynne (2017). *Theatre Aurality*. Palgrave Macmillan. [10.1057/978-1-137-45233-7](https://doi.org/10.1057/978-1-137-45233-7).
- Krautter, Benjamin and Janis Pagel (2024). "The Schemer in German Drama: Identification and Quantitative Characterization". In: *Computational Drama Analysis*. Ed. by Melanie Andresen and Nils Reiter. De Gruyter, 123–148. [10.1515/9783111071824-007](https://doi.org/10.1515/9783111071824-007).
- Krautter, Benjamin, Janis Pagel, Nils Reiter, and Marcus Willand (2020). "»[E]in Vater, dächte ich, ist doch immer ein Vater«. Figurentypen und ihre Operationalisierung". In: *Zeitschrift für digitale Geisteswissenschaften* 5. [10.17175/2020_007](https://doi.org/10.17175/2020_007).

- Krautter, Benjamin, Janis Pagel, Nils Reiter, and Marcus Willand (2022). "Properties of Dramatic Characters: Automatically Detecting Gender, Age, and Social Status". In: *Computational Stylistics in Poetry, Prose, and Drama*. De Gruyter, 179–202. [10.1515/9783110781502-010](#).
- Lemke, Marc, Konrad Sperfeld, and Jochen Zöllner (forthcoming). "Introducing NTEE: An Easy to Use Tool to Enrich TEI Files with Entities Based on State of the Art Neural Networks". preprint.
- Matthews, R. A. J. (1993). "Neural Computation in Stylometry I: An Application to the Works of Shakespeare and Fletcher". In: *Literary and Linguistic Computing* 8 (4), 203–210. [10.1093/lc/8.4.203](#).
- Mildorf, Jarmila (2019). "Aural Worldmaking: Introduction". In: *CounterText* 5 (3), 290–293. [10.3366/count.2019.0166](#).
- Molyneux, Frances (2022). *Sound's Effects: Sonic Epistemology in the Victorian Novel*. Stanford University.
- Montani, Ines, Matthew Honnibal, Adriane Boyd, Sofie Van Landeghem, and Henning Peters (2022). *spaCy: Industrial-strength Natural Language Processing in Python*. <https://spacy.io/> (visited on 03/06/2024).
- Pfister, Manfred (2001). *Das Drama. Theorie und Analyse*. München: Fink.
- Picker, John M. (2003). *Victorian Soundscapes*. Oxford University Press.
- Reiter, Nils (2021). "Möglichkeiten Quantitativer Dramenanalyse". In: *Comparatio* 12 (2), 39–52. <https://comp.winter-verlag.de/article/COMP/2020/2/7> (visited on 10/09/2021).
- Reiter, Nils and Marcus Willand (2023). "What are They Talking about? A Systematic Exploration of Theme Identification Methods for Character Speech in Dramatic Texts". In: *Digitale Literaturwissenschaft*. Ed. by Fotis Jannidis. Germanistische Symposien. J.B. Metzler, 473–508. [10.1007/978-3-476-05886-7_20](#).
- Schumacher, Mareike, Inna Uglanova, and Evelyn Gius (2022). *d-Romane-Romantik (d-RoRo)*. Version 1.0. Zenodo. [10.5281/zenodo.7223544](#).
- Smith, Bruce R. (1999). *The Acoustic World of Early Modern England: Attending to the O-factor*. University of Chicago Press.
- Szemes, Botond and Mihaly Nagy (2025). "Temporal Aspects of Structural Differences in Dramatic Genre". In: *Zeitschrift für digitale Geisteswissenschaften* 10. [10.17175/2025_007](#).
- Trilcke, Peer, Christopher Kittel, Nils Reiter, Daria Maximova, and Frank Fischer (2020). "Opening the Stage: A Quantitative Look at Stage Directions in German Drama". In: *Book of Abstracts of the DH2020*. https://dh2020.adho.org/wp-content/uploads/2020/07/337_OpeningtheStageAQuantitativeLookatStageDirectionsinGermanDrama.html (visited on 06/16/2025).
- Trilcke, Peer, Evgeniya Ustinova, Ingo Börner, Frank Fischer, and Carsten Milling (2024). "Detecting Small Worlds in a Corpus of Thousands of Theater Plays". In: *Computational Drama Analysis*. Ed. by Melanie Andresen and Nils Reiter. De Gruyter, 7–34. [10.1515/978311071824-002](#).
- Vishnubhotla, Krishnapriya, Adam Hammond, and Graeme Hirst (2019). "Are Fictional Voices Distinguishable? Classifying Character Voices in Modern Drama". In: *Proceedings of the 3rd Joint SIGHUM Workshop on Computational Linguistics for Cultural Heritage, Social Sciences, Humanities and Literature*. Minneapolis, USA: Association for Computational Linguistics, 29–34. <http://www.aclweb.org/anthology/W19-2504>.

- Weitin, Thomas (2021). *Digitale Literaturgeschichte: eine Versuchsreihe mit sieben Experimenten*. J.B. Metzler Series: Digitale Literaturwissenschaft. Springer/Nature.
- Zöllner, Jochen, Konrad Sperfeld, Christoph Wick, and Roger Labahn (2021). "Optimizing Small BERTs Trained for German NER". In: *Information* 12 (11), 443. [10.3390/info12110443](#).

A. Appendix

Experiment	Tok. Acc	Token Macro F1		
	F1	P	R	F1
<i>Original plays (gold = 1,747)</i>				
Prose (orig)	0.97 0.97	0.54 0.56	0.33 0.34	0.39 0.40
<i>Augmented plays (gold = 1,828)</i>				
Prose (aug)	0.97 0.97	0.57 0.61	0.29 0.30	0.37 0.38
Exp. o'	0.97 0.97	0.53 0.54	0.26 0.27	0.33 0.34
Exp. 2'	0.97 0.97	0.60 0.62	0.28 0.29	0.37 0.37
Exp. 3'	0.97 0.97	0.56 0.58	0.36 0.37	0.41 0.42
Exp. 4'	0.97 0.97	0.59 0.61	0.39 0.39	0.44 0.45
Exp. 1'	0.97 0.97	0.57 0.59	0.34 0.34	0.40 0.41

Table 6: Token accuracy and token-level macro F1, pooled over all twelve test plays; each cell reports before | after post-processing. Experiment labels: *Prose (orig)* and *Prose (aug)* apply the 64-prose model to the original and augmented plays; Exp. o' trains on the 8 augmented plays alone, and Exp. 2' / 3' / 4' / 1' add 8 / 16 / 32 / 64 prose texts. The two blocks are scored against different numbers of gold targets and are not directly comparable.

Experiment	Token F1 [ambient]			Token F1 [character]		
	P	R	F1	P	R	F1
<i>Original plays (gold = 1,747)</i>						
Prose (orig)	0.45 0.45	0.31 0.32	0.34 0.35	0.62 0.66	0.35 0.36	0.44 0.46
<i>Augmented plays (gold = 1,828)</i>						
Prose (aug)	0.50 0.52	0.21 0.22	0.27 0.28	0.63 0.69	0.38 0.38	0.46 0.48
Exp. o'	0.42 0.43	0.13 0.14	0.19 0.19	0.64 0.65	0.39 0.39	0.48 0.48
Exp. 2'	0.49 0.52	0.16 0.17	0.23 0.23	0.72 0.73	0.40 0.41	0.51 0.51
Exp. 3'	0.47 0.48	0.23 0.24	0.28 0.28	0.66 0.69	0.48 0.49	0.55 0.56
Exp. 4'	0.52 0.52	0.33 0.34	0.36 0.36	0.65 0.69	0.45 0.45	0.51 0.53
Exp. 1'	0.47 0.47	0.23 0.24	0.29 0.30	0.66 0.70	0.44 0.45	0.52 0.53

Table 7: Token-level F1 by sound class, pooled over all twelve test plays; before | after post-processing. Configuration as in Table 6.

Experiment	Span det. (any)			Span det. (IoU $\geq$.5)		
	P	R	F1	P	R	F1
<i>Original plays (gold = 1,747)</i>						
Prose (orig)	0.58 0.71	0.43 0.43	0.48 0.52	0.50 0.64	0.37 0.39	0.41 0.47
<i>Augmented plays (gold = 1,828)</i>						
Prose (aug)	0.57 0.71	0.54 0.51	0.53 0.58	0.45 0.59	0.42 0.43	0.42 0.49
Exp. o'	0.58 0.64	0.56 0.54	0.56 0.58	0.46 0.52	0.44 0.44	0.45 0.47
Exp. 2'	0.64 0.71	0.57 0.54	0.59 0.60	0.51 0.61	0.45 0.46	0.47 0.52
Exp. 3'	0.60 0.70	0.59 0.58	0.59 0.62	0.51 0.61	0.50 0.51	0.50 0.55
Exp. 4'	0.59 0.69	0.57 0.55	0.57 0.60	0.50 0.62	0.49 0.50	0.49 0.55
Exp. 1'	0.63 0.73	0.58 0.57	0.59 0.62	0.53 0.64	0.49 0.50	0.50 0.55

Table 8: Span detection (label ignored), pooled over all twelve test plays; before | after post-processing. “any” uses any-overlap matching, “IoU $\geq$.5” at least half-overlap with the gold span. Configuration as in Table 6.

Experiment	Span class. (any)			Span class. (IoU $\geq$.5)		
	P	R	F1	P	R	F1
<i>Original plays (gold = 1,747)</i>						
Prose (orig)	0.56 0.68	0.42 0.41	0.46 0.50	0.48 0.62	0.36 0.37	0.40 0.45
<i>Augmented plays (gold = 1,828)</i>						
Prose (aug)	0.55 0.68	0.51 0.49	0.51 0.56	0.43 0.57	0.40 0.41	0.40 0.46
Exp. o'	0.56 0.61	0.54 0.52	0.54 0.56	0.45 0.50	0.43 0.43	0.43 0.46
Exp. 2'	0.62 0.68	0.55 0.52	0.57 0.58	0.49 0.59	0.43 0.45	0.45 0.50
Exp. 3'	0.58 0.67	0.57 0.56	0.56 0.60	0.49 0.59	0.49 0.50	0.48 0.53
Exp. 4'	0.57 0.66	0.55 0.54	0.55 0.58	0.48 0.59	0.47 0.48	0.47 0.52
Exp. 1'	0.60 0.70	0.55 0.54	0.56 0.60	0.51 0.62	0.47 0.48	0.48 0.53

Table 9: Span classification (correct label required), pooled over all twelve test plays; before | after post-processing. Matching criteria as in Table 8. Configuration as in Table 6.

Experiment	E-F1 (exact span + label)			Gamma
	P	R	F1	F1
<i>Original plays (gold = 1,747)</i>				
Prose (orig)	0.42 0.55	0.31 0.33	0.35 0.40	0.43 0.48
<i>Augmented plays (gold = 1,828)</i>				
Prose (aug)	0.34 0.45	0.32 0.32	0.32 0.37	0.45 0.51
Exp. o'	0.39 0.43	0.38 0.37	0.38 0.39	0.48 0.51
Exp. 2'	0.43 0.51	0.37 0.39	0.39 0.43	0.51 0.54
Exp. 3'	0.44 0.53	0.43 0.44	0.43 0.47	0.53 0.57
Exp. 4'	0.40 0.50	0.40 0.41	0.39 0.44	0.51 0.55
Exp. 1'	0.44 0.53	0.40 0.42	0.41 0.46	0.52 0.57

Table 10: Overall entity-level F1 (E-F1) and Gamma agreement (Mathet et al.), pooled over all twelve test plays; before | after post-processing. Configuration as in Table 6.

Experiment	E-F1 [ambient]			E-F1 [character]		
	P	R	F1	P	R	F1
<i>Original plays (gold = 1,747)</i>						
Prose (orig)	0.26 0.32	0.25 0.29	0.23 0.28	0.44 0.58	0.31 0.33	0.35 0.41
<i>Augmented plays (gold = 1,828)</i>						
Prose (aug)	0.13 0.19	0.10 0.12	0.11 0.14	0.36 0.48	0.35 0.36	0.34 0.40
Exp. 0'	0.17 0.20	0.08 0.08	0.10 0.11	0.42 0.45	0.42 0.41	0.42 0.42
Exp. 2'	0.16 0.24	0.08 0.10	0.10 0.13	0.46 0.55	0.42 0.44	0.43 0.48
Exp. 3'	0.22 0.28	0.15 0.18	0.16 0.19	0.47 0.55	0.48 0.49	0.47 0.51
Exp. 4'	0.19 0.21	0.19 0.21	0.18 0.21	0.43 0.53	0.42 0.43	0.42 0.46
Exp. 1'	0.17 0.21	0.16 0.18	0.16 0.19	0.47 0.56	0.44 0.45	0.44 0.49

Table 11: Entity-level F1 (E-F1) by sound class, pooled over all twelve test plays; before | after post-processing. Configuration as in Table 6.

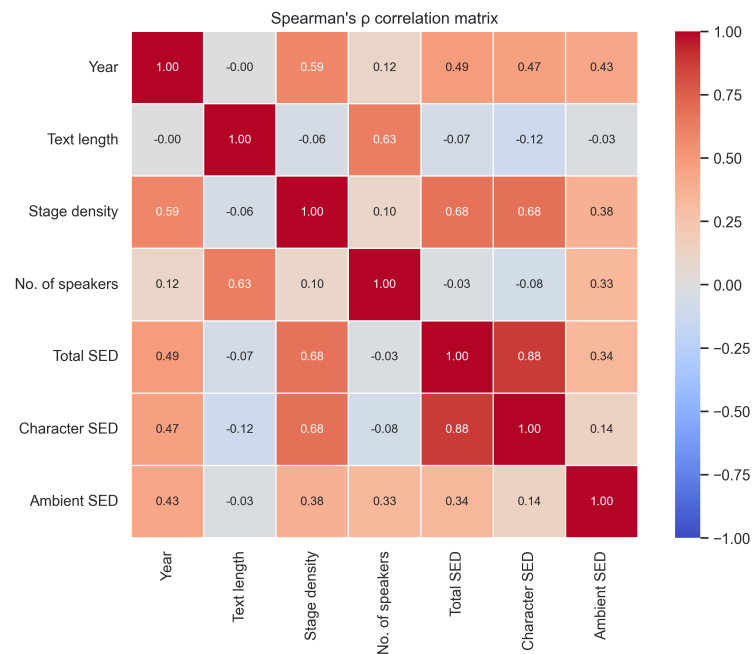


Figure 9: Spearman's ρ correlation matrix for the seven key variables across the sound-annotated GerDraCor plays. Stage direction density shows the strongest correlation with Total SE_D ($\rho = 0.68$) and Character SE_D ($\rho = 0.68$), while text length and number of speakers are largely uncorrelated with the SE_D measures. Notably, text length correlates strongly with number of speakers ($\rho = 0.63$), reflecting the tendency of longer plays to feature more characters, but neither variable shows a meaningful relationship with sound event density. The weaker correlation between stage direction density and Ambient SE_D ($\rho = 0.38$) compared to Character SE_D suggests that the two sound types respond differently to the expansion of stage directions over time.